Preface

The challenge for the engineer and technician today is to make effective use of modern instrumentation and control systems and ‘smart’ instruments. This is achieved by linking equipment such as PCs, programmable logic controllers (PLCs), SCADA and distributed control systems, and simple instruments together with data communications systems that are correctly designed and implemented. In other words: *to fully utilize available technology*.

Practical Data Communications for Instrumentation and Control is a comprehensive book covering industrial data communications including RS-232, RS-422, RS-485, industrial protocols, industrial networks, and communication requirements for ‘smart’ instrumentation.

Once you have studied this book, you will be able to analyze, specify, and debug data communications systems in the instrumentation and control environment, with much of the material presented being derived from many years of experience of the authors. It is especially suited to those who work in an industrial environment and who have little previous experience in data communications and networking.

Typical people who will find this book useful include:

- Instrumentation and control engineers and technicians
- Process control engineers and technicians
- Electrical engineers
- Consulting engineers
- Process development engineers
- Design engineers
- Control systems sales engineers
- Maintenance supervisors

We would hope that you will gain the following from this book:

- The fundamentals of industrial data communications
- How to troubleshoot RS-232 and RS-485 links
- How to install communications cables
- The essentials of industrial Ethernet and local area networks
- How to troubleshoot industrial protocols such as Modbus
- The essentials of Fieldbus and DeviceNet standards

You should have a modicum of electrical knowledge and some exposure to industrial automation systems to derive maximum benefit from this book.

Why do we use RS-232, RS-422, RS-485 ?

One is often criticized for using these terms of reference, since in reality they are obsolete. However, if we briefly examine the history of the organization that defined these standards, it is not difficult to see why they are still in use today, and will probably continue as such.

The common serial interface RS-232 was defined by the Electronics Industry Association (EIA) of America. ‘RS’ stands for Recommended Standards, and the number (suffix -232) refers to the interface specification of the physical device. The EIA has since established many standards and amassed a library of white papers on various implementations of them. So to keep track of

them all it made sense to change the prefix to EIA. (You might find it interesting to know that most of the white papers are NOT free).

The Telecommunications Industry Association (TIA) was formed in 1988, by merging the telecom arms of the EIA and the United States Telecommunications Suppliers Association. The prefix changed again to EIA/TIA-232, (along with all the other serial implementations of course). So now we have TIA-232, TIA-485 etc.

We should also point out that the TIA is a member of the Electronics Industries Alliance (EIA). The alliance is made up of several trade organizations (including the CEA, ECA, GEIA...) that represent the interests of manufacturers of electronics-related products. When someone refers to 'EIA' they are talking about the Alliance, not the Association!

If we still use the terms EIA-232, EIA-422 etc, then they are just as equally obsolete as the 'RS' equivalents. However, when they are referred to as TIA standards some people might give you a quizzical look and ask you to explain yourself... So to cut a long story short, one says 'RS-xxx' and the penny drops.

In the book you are about to read, the authors have painstakingly altered all references for serial interfaces to 'RS-xxx', after being told to change them BACK from 'EIA-xxx'! So from now on, we will continue to use the former terminology. This is a sensible idea, and we trust we are all in agreement!

Why do we use DB-25, DB-9, DB-xx ?

Originally developed by Cannon for military use, the D-sub(miniature) connectors are so-called because the shape of the housing's mating face is like a 'D'. The connectors have 9-, 15-, 25-, 37- and 50-pin configurations, designated DE-9, DA-15, DB-25, DC-37 and DD-50, respectively. Probably the most common connector in the early days was the 25-pin configuration (which has been around for about 40 years), because it permitted use of all available wiring options for the RS-232 interface.

It was expected that RS-232 might be used for synchronous data communications, requiring a timing signal, and thus the extra pin-outs. However this is rarely used in practice, so the smaller 9-position connectors have taken its place as the dominant configuration (for asynchronous serial communications).

Also available in the standard D-sub configurations are a series of high density options with 15-, 26-, 44-, and 62-pin positions. (Possibly there are more, and are usually variations on the original A,B,C,D, or E connector sizes). It is common practice for electronics manufacturers to denote all D-sub connectors with the DB- prefix... particularly for producers of components or board-level products and cables. This has spawned generations of electronics enthusiasts and corporations alike, who refer to the humble D-sub or 'D Connector' in this fashion. It is for this reason alone that we continue the trend for the benefit of the majority who are so familiar with the 'DB' terminology.

The structure of the book is as follows.

Chapter 1: Overview. This chapter gives a brief overview of what is covered in the book with an outline of the essentials and a historical background to industrial data communications.

Chapter 2: Basic principles. The aim of this chapter is to lay the groundwork for the more detailed information presented in the following chapters.

Chapter 3: Serial communication standards. This chapter discusses the main physical interface standards associated with data communications for instrumentation and control systems.

Chapter 4: Error detection. This chapter looks at how errors are produced and the types of error detection, control, and correction available.

Chapter 5: Cabling basics. This chapter discusses the issues in obtaining the best performance from a communication cable by selecting the correct type and size.

Chapter 6: Electrical noise and interference. This chapter examines the various categories of electrical noise and where each of the various noise reduction techniques applies.

Chapter 7: Modems and multiplexers. This chapter reviews the concepts of modems and multiplexers, their practical use, position and importance in the operation of a data communication system.

Chapter 8: Introduction to protocols. This chapter discusses the concept of a protocol which is defined as a set of rules governing the exchange of data between a transmitter and receiver over a communications link or network.

Chapter 9: Open systems interconnection model. The purpose of the Open Systems Interconnection reference model is to provide a common basis for the development of systems interconnection standards. An open system is a system that conforms to specifications and guidelines, which are 'open' to all.

Chapter 10: Industrial protocols. This chapter focusses on the software aspects of protocols (as opposed to the physical aspects which are covered in earlier chapters).

Chapter 11: HART protocol. The Highway Addressable Remote Transducer (HART) protocol is one of a number of smart instrumentation protocols designed for collecting data from instruments, sensors and actuators by digital communication techniques. This chapter examines this in some depth.

Chapter 12: Open industrial Fieldbus and DeviceNet systems. This chapter examines the different Fieldbus and DeviceNet systems on the market with an emphasis on ASI Bus, CanBus and DeviceNet, Interbus-S, Profibus and Foundation Fieldbus.

Chapter 13: Local area networks (LANs). This chapter focuses on networks generally used in industrial data communications with an emphasis on Ethernet.